

Practice Final B

Score

MTH 207 (Fall 2026) — J. Baggett

/ 235

During the exam:

- **TO EARN POINTS YOU MUST SHOW YOUR WORK:**
 - Explain your reasons and justify your steps.
 - Demonstrate that you understand the *process*.
 - Present your logic in an organized manner on paper.
- Keep your eyes on your own exam!
- No calculators, phones, or other electronics at all. They should be turned off!
- No notes, books, or other outside resources.

The real final is 120 minutes and 200 points. This practice final is longer (220 points) so that it covers more of the material — do not use it to judge timing.

- When you are told to stop, you must stop **IMMEDIATELY**. This is in fairness to all students. Do not think that you are the exception to this rule. Continuing to work on the exam after time has been called will have serious consequences.

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(3^{pts} ea.) **1.** Determine whether each statement is **true** or **false**. If it is true, say why in one sentence. If it is false, explain why or give an example that disproves it.

(a) If $f(2) = 5$, then $\lim_{x \rightarrow 2} f(x) = 5$.

(b) If $s(t)$ is position, then $s'(a)$ is the instantaneous velocity at time a , and it can be negative.

(c) If two quantities are related by an equation, then their rates of change are equal.

(d) If f is periodic, then f' is periodic.

(e) If f is continuous on $[a, b]$, then $\frac{d}{dx} \left(\int_a^b f(x) dx \right) = f(x)$.

(7^{pts} ea.) **2.** Evaluate the following limits. As a reminder, justify your steps! Simplify your answer as much as possible. (If the limit does not exist, state why.)

(a) $\lim_{x \rightarrow \infty} \frac{5x + 4}{3x - 7}$

(b) $\lim_{x \rightarrow \infty} (\sqrt{9x^2 + 2x} - 3x)$

(c) $\lim_{x \rightarrow \infty} \left(1 + \frac{3}{x}\right)^x$

- (10^{pts}) **3.** Sketch the region enclosed by the curves $y = 6 - x^2$ and $y = x$, showing the points where they intersect. Then set up and evaluate an integral for its area.
- (3^{pts}) **4.** Sketch the graph of a function that is continuous everywhere but fails to be differentiable at exactly two x -values. Label those two x -values on your sketch, and say in one sentence what goes wrong at each.
- (20^{pts}) **5.** A streetlight is mounted at the top of a 15-foot pole. A person 6 feet tall walks away from the pole along a straight path at a rate of 5 feet per second. How fast is the tip of the person's shadow moving along the ground when the person is 40 feet from the pole? (Be sure to include units.)

(8^{pts} ea.) **6.** Evaluate the following indefinite integrals. Simplify your answer as much as possible.

(a) $\int \frac{x^3 + 2x^2 - x}{x} dx$

(b) $\int \frac{3^{\cos x}}{\csc x} dx$

(c) $\int \frac{\sin(2 + \ln x)}{x} dx$

- (12^{pts}) **7.** The point $(3, 1)$ lies on the curve $x^2y + y^3 = 10$. Find the slope of the curve at that point, and then determine whether the tangent line there passes through the origin. Justify your conclusion.

- (10^{pts}) **8.** Given a function $f(x)$, write a formula for the average rate of change of f on the interval $[a, a + h]$. Explain what this quantity measures, and explain how the derivative $f'(a)$ is obtained from it.

(5^{pts} ea.) **9.** Evaluate the following limits. Justify your steps, and name any rule you use.

(a) $\lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2x}{x^2}$

(b) $\lim_{x \rightarrow 0^+} x \ln x$

10^{pts} ea.) **10.** Evaluate the following definite integrals. Simplify your answer as much as possible.

(a) $\int_0^1 \frac{x}{1+x^2} dx$

(b) $\int_0^{\pi/4} \frac{2+\cos^2 \theta}{\cos^2 \theta} d\theta$

(c) $\int_{-3}^{3\sqrt{3}} \frac{2}{\sqrt{36-x^2}} dx$

(8^{pts}) **11.** Given a fixed number $a > 0$ with $a \neq 1$, derive the derivative rule for $y = \log_a x$. (You may use the derivative rule for $\ln x$.)

(15^{pts}) **12.** Use the definition of the definite integral as a limit of Riemann sums to show that $\int_0^3 (4x - 1) dx = 15$. Then confirm your answer using the Fundamental Theorem of Calculus. (You may use any sampling rule, and you may use $\sum_{i=1}^n i = \frac{n(n+1)}{2}$.)

(10^{pts}) **13.** Suppose f is continuous on $(-\infty, \infty)$ and satisfies all of the following:

$$f'(x) > 0 \text{ on } (-\infty, 1) \text{ and } (3, \infty), \quad f'(x) < 0 \text{ on } (1, 3),$$

$$f''(x) < 0 \text{ on } (-\infty, 2), \quad f''(x) > 0 \text{ on } (2, \infty).$$

Identify each critical number of f and classify it as a local maximum, a local minimum, or neither, giving your reason. State the x -coordinate of every inflection point. Then sketch a graph that could be f .

(15^{pts}) **14.** Find y' if $y = 2^x \cos x + \ln(\sin^{-1} x) + 3 \int_x^7 \sqrt{1+t^4} dt$.

- (2^{pts}) **15.** Consider $f(x) = \sqrt{x-1}$ on $[5, 17]$. According to the Mean Value Theorem there is a number c in $(5, 17)$ so that $f'(c) = \underline{\hspace{2cm}}$. Compute and simplify the number to fill in the blank.

- (10^{pts}) **16.** Choose ONE integral on this page and evaluate it. Clearly mark which integral is your choice!
(Suggestion: do your scratch work on the back page or ask for extra paper.)

$$\int \frac{e^x}{1 + e^{2x}} dx$$

$$\int \frac{dx}{\sqrt{x}(1+x)}$$

$$\int_0^4 x \sqrt{4 - (x-2)^2} dx$$

$$\int \frac{dx}{\sqrt{-x^2 + 4x}}$$

- (20^{pts}) **17.** A bus company will charge \$200 per person for a tour if 50 people sign up. For each additional person beyond 50 who signs up, the company reduces the price by \$2 per person for everyone on the tour. How many additional people should sign up in order to maximize the company's total revenue?

THIS IS THE LAST PAGE OF THE EXAM.